

# Lecture 13 and 14

- consumer & producer surplus, and deadweight loss
  - I said I might skip this but decided to do briefly
- double auctions
- exchange economies

# Consumer & Producer Surplus

- consumer surplus (CS) is the area under demand, above equilibrium price, left of quantity traded
  - how much surplus are consumers getting (value - actual price, integrated across all consumers)
  - mathematically,  $\int_0^{q^*} (P^D(q) - p^*) dq$
  - this is often used as a measure of consumer welfare, and so the welfare impact of taxes can be measured by its change
- producer surplus (PS) is the area above supply, below equilibrium price, left of quantity traded
  - how much surplus are suppliers getting (price - amount they would have been willing to accept, integrated across all consumers)
  - mathematically,  $\int_0^{q^*} (p^* - P^S(q)) dq$

## Example

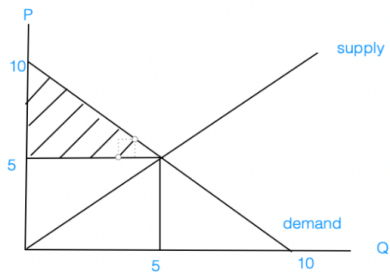
- assume demand  $P^D(Q) = 10 - Q$  and supply  $P^S(Q) = Q$
- equating supply and demand, we find equilibrium quantity  $Q^* = 5$  and equilibrium price  $P^* = 5$
- using the integral formulas,

$$CS = \int_0^{Q^*} (P^D(Q) - p^*) dQ = \int_0^5 (10 - q - 5) dq = 5q - \frac{1}{2}q^2 \Big|_{q=0}^{q=5}$$

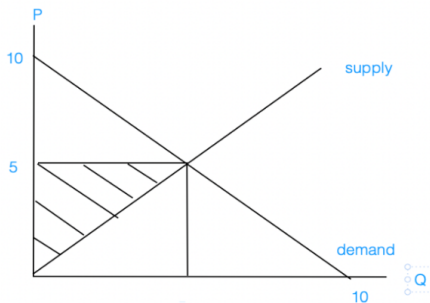
$$PS = \int_0^{Q^*} (p^* - P^S(Q)) dQ = \int_0^5 (5 - Q) dQ = 5q - \frac{1}{2}q^2 \Big|_{q=0}^{q=5}$$

- but with linear supply and demand, if you don't like integrals, you can use high school geometry!

# Example: Consumer Surplus



# Example: Producer Surplus



# Deadweight Loss

- the *deadweight loss* of a tax is the loss in surplus  $CS+PS$  + government revenue due to quantity distortions
- to illustrate in our example above, consider a \$1 tax, say on consumers (recall it doesn't matter)
- this shifts demand down to  $P^D(Q) = 9 - Q$  (we learned this before the midterm), while supply is again  $P^S(Q)$
- post-tax equilibrium:  $Q^* = P^* = 4.5$  in the market, so suppliers get \$4.5 and consumers pay \$5.5 with tax
- recall that pre-tax,  $CS=PS=12.5$

## Deadweight Loss, Cont'd

- post-tax CS:

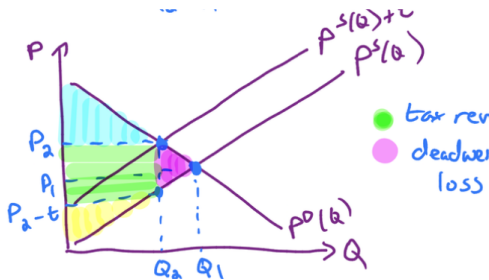
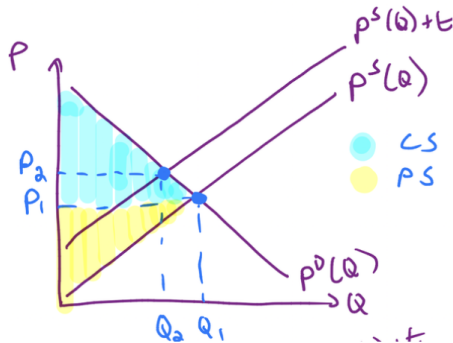
$$\int_0^{q^*} (P^D(q) - p^*) dq = \int_0^{4.5} (10 - q - 5.5) dq = 10.125$$

- post-tax PS:

$$\int_0^{q^*} (p^* - P^S(q)) ds = \int_0^{4.5} (4.5 - q) dq = 10.125$$

- government revenue:  $Q \cdot t = (4.5)(1) = 4.5$
- new total surplus:  $10.125 + 10.125 + 4.5 = 24.75$
- old total surplus:  $12.5 + 12.5 = 25$
- the 0.25 unit drop is the deadweight loss

# Deadweight Loss



# Market Exchange: Double Auctions

- consider a world with a fixed finite # buyers, and a fixed finite # sellers, trading **indivisible** goods
  - NOTE: this is the one part of the course where you can't buy decimal quantities!
- in this world, a *competitive equilibrium* is a set of prices and a trading allocation in which all agents are optimizing (given prices)
- namely, no trading agent could earn greater profits from a different (or non-) trade, and no non-trading agent can profitably trade

# Double Auctions: The Details

- each buyer  $j$  wants to buy at most one unit, which he values at  $v_j$
- each seller  $i$  has one unit to sell, with cost (or equivalently, value)  $c_i$
- the one unit assumption is not important, but it is key that goods are **indivisible**
- *quasilinear utility*: if buyer  $j$  and seller  $i$  trade at price  $p$ ,
  - buyer payoff:  $v_j - p$
  - seller payoff:  $p - c_i$
  - total trade surplus:  $v_j - c_i$
- suppose goods are allocated via *double auction*: buyers and sellers simultaneously submit their bids for the goods
- what will the equilibrium look like?

# Equilibrium Feature #1: One Price for All

- all trades must happen at the same price  $p$
- proof: suppose not, for example say one buyer-seller pair trades for \$3 and the other for \$5
- then the second buyer (who pays \$5 and would prefer less) and the first buyer (who sells for \$3 and would prefer more) can strike up a profitable “side deal”, eg trade at \$4
- equilibrium  $\Rightarrow$  no profitable side deals should be possible

## Feature #2: Highest-Value Buyers, Lowest-Cost Sellers Trade

- by feature #1, all trade at the same price, let's say it's  $p$
- it can't be that a low-value ( $v$ ) buyer gets the good, but a high-value ( $v' > v$ ) buyer doesn't
  - since the low-value buyer is willing to buy, it must be that  $p \leq v$
  - but then  $p < v'$ , so the high-value buyer would strictly like to purchase the good, and can't get one
  - $\Rightarrow$  excess demand, not an equilibrium
- similarly, can't be that a high-cost ( $c'$ ) seller sells the good, but a low-cost ( $c < c'$ ) seller does not
  - high-cost seller would only sell if  $p \geq c'$
  - but then  $p > c$ , so the low-cost seller strictly wants to sell, and cannot
  - $\Rightarrow$  excess supply, not an equilibrium!

## Operationally: How to Apply This?

- order the sellers from lowest-to-highest and order the buyers highest-to-lowest
- “clear the market” left to right, meaning, match up each pair with a positive trade surplus
  - this determines the equilibrium quantity, let's say it's  $q^*$
- now, choose a price that makes this an equilibrium:
  - first, the price must encourage trade  $q^*$ :  $c_{q^*} \leq p \leq v_{q^*}$  (ensures all trading pairs are happy to trade)
  - second, the trade must discourage trade  $q^* + 1$ :  $c_{q^*+1} \geq p \geq v_{q^*+1}$  (ensures all non-trading pairs are happy not to trade)

# Competitive Equilibrium: Features and Notes

- in our discrete world, the competitive equilibrium is typically not unique
- each equilibrium maximizes total surplus, and is immune to side bribes
- note that since we now have an integer restriction, find CS and PS by adding, not integrating
  - CS: add up  $v_j - p^*$  across all trading buyers  $j$
  - PS: add up  $p^* - c_i$  across all trading sellers  $i$
  - CS+PS is then the total trade surplus

## Example 1

- Consider 20 agents, numbered from 1 to 20
- The even agents are buyers, and the odd agents are sellers
- buyer  $j$ 's valuation is  $v_j = 2j$ , and seller  $i$ 's cost is  $c_i = 3i$

## Example 1

- ordering the buyer valuations from high to low, and seller costs low to high, we get:

|              |    |    |    |    |    |    |    |    |    |    |
|--------------|----|----|----|----|----|----|----|----|----|----|
| buyer values | 40 | 36 | 32 | 28 | 24 | 20 | 16 | 12 | 8  | 4  |
| seller costs | 3  | 9  | 15 | 21 | 27 | 33 | 39 | 45 | 51 | 57 |

- to find an efficient matching, clear the market “left to right”, i.e. start pairing the highest-valuation buyers with the lowest-cost sellers, finding equilibrium trading quantity  $q^* = 4$  (beyond this, trade surplus is negative)
- for a competitive equilibrium, we need a price  $p^*$  such that the 4th highest buyer + seller do want to trade,

$$21 \leq p^* \leq 28$$

but the 5th pair do not:

$$24 \leq p^* \leq 27$$

## Example 1

- the 2nd condition is the more restrictive one, so we conclude that a competitive equilibrium is any pair  $(p^*, q^*)$  with  $p^* \in [24, 27]$  and  $q^* = 4$
- this guarantees that every buyer who buys, and every seller who sells, is happy to do so
  - eg, the 4th buyer gets a good worth 28 for a price  $p^* \in [24, 27]$ , while 4th seller produces a good for \$21 and earns  $p^* \in [24, 27]$  for it
- this also guarantees that every buyer/seller who's left out is happy to be left out (no excess demand/supply)
  - all buyers left out have valuations  $\leq 24$ , and the price is  $p^* \geq 24$
  - all sellers left out have costs  $\geq 27$ , and the price is  $p^* \leq 27$

## Example, Modified

- now suppose there are 25 agents, numbered from 1-25
- the even agents are buyers, with valuation  $3j$  for buyer  $j$
- the odd agents are sellers, with valuation  $2i$  for seller  $i$
- ordering the buyer valuations high-low, and seller costs low-high,

|              |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|
| buyer values | 72 | 66 | 60 | 54 | 48 | 42 | 36 | 30 | 24 | 18 | 12 | 6  |    |
| seller costs | 2  | 6  | 10 | 14 | 18 | 22 | 26 | 30 | 34 | 38 | 42 | 46 | 50 |

- In this case the unique equilibrium price is  $p^* = 30$  and the equilibrium quantity is  $q^* = 7$  or  $q^* = 8$ .

## Another Example

- consider a continuum of buyers  $b \in [0, 1]$ , where buyer  $b$  has valuation  $b$  and is willing to buy 1 unit
- and a continuum of sellers  $s \in [0, 1]$ , where seller  $s$  has valuation  $s$  and is willing to sell 2 units
- find the equilibrium price and quantity

# Solution

- equilibrium needs demand = supply, so let's derive demand and supply curves
- demand: buyer  $b$  will only buy at price  $p$  if  $b \geq p$ , in which case he wants 1 unit: so demand is

$$Q^D(p) = \int_p^1 1db = 1 - p$$

- supply: seller  $s$  will only sell at price  $p$  if  $s \leq p$ , in which case he wants 2 units: so supply is

$$Q^S(p) = \int_0^p 2ds = 2p$$

- equilibrium price equates supply = demand, so  $1 - p = 2p \Rightarrow p = \frac{1}{3}$
- at this point, supply and demand quantity are both  $\frac{2}{3}$

# Exchange Economies

- an exchange economy is one where there is no production, and no outside money
- rather, each has an endowment of goods, which is his “wealth”
- we'll focus on 2 people, 2 goods
- the two people can trade with each other if they wish
- but instead of asking them to directly trade goods, we'll facilitate it by setting up a market:
  - specify prices at which goods can be *bought AND sold*
  - your wealth is the cash value (at these prices) of your endowment bundle

# Big Ideas

- first, we'll formalize what an *equilibrium* is, and how to find it
  - basically a simpler version of our usual market equilibrium
  - nobody thinks they can do better, and supply=demand
- then we'll learn how to find **Pareto optimal allocations**
  - this basically asks, suppose there were a benevolent social planner dividing up resources
  - what are the most efficient ways to do so?
- finally, explore the connection between these concepts

# Big Ideas

- 1st and 2nd welfare theorems (apply in more general markets too) basically say
- If we let people optimize as they wish, they'll do something efficient (1st welfare theorem)
  - so the only scope for government intervention is to redistribute resources differently
- But (2nd welfare theorem) any efficient redistributive goal can be achieved via taxes and subsidies, while leaving people free choice in what they consume

# Equilibrium

- in equilibrium, everyone should feel they're doing the best they can do
- this imposes two requirements
- **individual optimization:** each person gets their best affordable bundle
  - their “wealth” is the cash value of their endowment
  - so they can afford any bundle whose value does not exceed wealth
- **market-clearing:** demand = supply for each good
  - otherwise, if eg there's excess demand, it means that at the prevailing prices, someone wants to buy more goods than are available; this leaves room for profitable side deals, not an equilibrium

## Example

- Marge and Homer are stranded on an island.
- Endowments:
  - Homer has 12 bottles of beer and 0 donuts
  - Marge has 4 bottles of beer and 8 donuts.
- Market: To help them trade, their daughter Lisa announces prices  $p_B, p_D$  at which they can buy AND sell beer and donuts.
- Preferences: both have  $u(B, D) = B^{\frac{3}{4}} D^{\frac{1}{4}}$
- Find the equilibrium price ratio  $p_B/p_D$  that clears the donut market, and double-check that it clears the beer market too

## Solution Part 1: Homer's Demand

- first, let us look at Homer's optimization problem
- he wants to choose  $B$  and  $D$  to maximize  $B^{\frac{3}{4}}D^{\frac{1}{4}}$ , subject to the *budget constraint for his endowment* ( $B=12, D=0$ ):

$$p_B B + p_D D \leq 12p_B + 0p_D$$

- recall from 1st half of course: for these Cobb-Douglas preferences, Homer optimally spends  $\frac{3}{4}$  of his wealth on beer and  $\frac{1}{4}$  on donuts
- so, his demand bundle is:

$$B^H = \frac{3}{4} \frac{12p_B + 0p_D}{p_B} = 9, \quad D^H = \frac{1}{4} \frac{12p_B + 0p_D}{p_D} = 3 \frac{p_B}{p_D}$$

## Solution Part 2: Marge's Demand

- next, let's look at Marge's optimal consumption problem
- she wants to choose  $B$  and  $D$  to maximize  $B^{\frac{3}{4}}D^{\frac{1}{4}}$ , subject to her budget constraint

$$p_B B + p_D D \leq 4p_B + 8p_D$$

- like Homer, her optimal bundle involves spending  $\frac{3}{4}$  of her income on beer, and  $\frac{1}{4}$  of her income on donuts:

$$B^M = \frac{3}{4} \frac{4p_B + 8p_D}{p_B} = 3 + 6 \frac{p_D}{p_B}, \quad D^M = \frac{1}{4} \frac{4p_B + 8p_D}{p_D} = \frac{p_B}{p_D} + 2$$

# Equilibrium in our Exchange Economy

- if we consider the donut market, we need total donut demand to equal total donut supply
- total donut supply is just 8 (since Marge has 8 and Homer has 0)
- total donut demand is

$$D^H + D^M = 3 \frac{p_B}{p_D} + 2 + \frac{p_B}{p_D} = 2 + 4 \frac{p_B}{p_D}$$

- so, our equilibrium condition is:

$$2 + 4 \frac{p_B}{p_D} = 8 \Rightarrow \frac{p_B}{p_D} = \frac{3}{2}$$

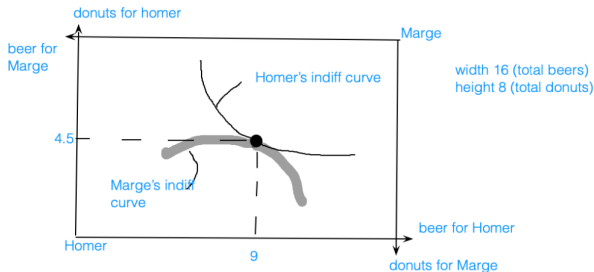
# Pareto Optimality

- **Def'n 1:** a *feasible allocation* is one where, for each good, total quantity demanded  $\leq$  total endowment
- **Def'n 2:** a feasible allocation is *Pareto optimal* if there's no feasible way to make one person better off without harming the other person
  - a Pareto improvement is a feasible change that helps both people (one strictly)
  - so an allocation is Pareto optimal if no Pareto improvements are possible

## Checking Pareto Optimality of our Exchange Eqm

- in our equilibrium, Marge and Homer both chose allocations where  $\frac{MU_B}{MU_D} = \frac{p_B}{p_D} = \frac{3}{2}$
- this means that they choose bundles where their indifference curves have the same slope
- so if you draw an *edgeworth box*, with width equal to total beer endowment 16, height total donut endowment 8, and indicate the allocation where Homer gets  $(B, D) = (9, 4.5)$  and Marge gets the rest, namely  $(B, D) = (7, 3.5)$ , their indifference curves are *tangent*
- it is then visually clear from the Edgeworth box that there is no way to make one of Marge/Homer better off without hurting the other, so our equilibrium is Pareto efficient

# Edgeworth Box



# Finding Pareto Optimal Allocations

- we said it was “visually obvious” that a tangency point in the Edgeworth box is Pareto optimal
- but in fact, with convex preferences, set of all Pareto optimal allocations = set of all tangency points in the Edgewroth box (why?)

## Finding Pareto Optimal Allocations

- in our above example,  $\frac{MU_B}{MU_D} = 3\frac{D}{B}$ , so the tangency condition is

$$3\frac{D^H}{B^H} = 3\frac{D^M}{B^M}$$

- and the feasibility condition is  $D^M = 8 - D^H$ ,  $B^M = 16 - B^H$
- substituting feasibility into tangency,
- Substituting the 2nd equations into the 1st for Marge's bundle,

$$3\frac{D^H}{B^H} = 3\frac{8 - D^H}{16 - B^H} \Rightarrow 2D^H = B^H$$

- so where Homer gets twice as many beers as donuts, and Marge gets everything else

# Pareto Optimality Without Convexity

- generally, this is tougher and doesn't have a standard recipe
- you need to use logic: Pareto efficient  $\Leftrightarrow$  no Pareto improvements possible
- if you cannot figure it out logically, you can also use this fact: an allocation is Pareto optimal if it uses up the aggregate endowment, and maximizes some weighted sum of utilities
  - set of all Pareto optimal allocations: all those that maximize  $\lambda u(B^H, D^H) + (1 - \lambda)u(B^M, D^M)$ , subject to using up entire endowment

# Non-Convex Examples

- example: both have  $u(B, D) = B + D$ 
  - *everything* in the Edgeworth box is Pareto optimal. For any point, both are willing to trade at 1-for-1 rate, leaving no mutually beneficial trades
- example: both have  $u(B, D) = \min\{B, D\}$  and aggregate endowments of B,D are equal
  - Pareto optimal: the main diagonal of the edgeworth box, where both have  $B=D$ . (Otherwise, one has extra beers but is short on donuts, and the other has extra donuts but is short on beers; both are helped by exchanging their “extra” good for the one they’re short on)
- example: both have concave preferences  $u(B, D) = B + D^2$ . Pareto optimal allocations are those where one person has all the donuts. Beers can be split in any way. (Not efficient for both to have positive quantities of donuts; whoever has more values them more, thus should get all of them).

# Equilibrium and Pareto Efficiency

## Theorem

*(1st Welfare Theorem): If preferences are locally non-satiated, then any competitive equilibrium is Pareto-efficient.*

- definition: preferences are locally non-satiated if for every point, any neighborhood around it contains something strictly preferred to the point

# Proof of 1st Welfare Theorem (for 2-person economies)

- suppose, toward a contradiction, that there is a feasible allocation  $x = (x_B^H, x_D^H, x_B^M, x_D^M)$  that is a competitive equilibrium, but isn't Pareto efficient
- since  $x$  isn't Pareto-efficient, there must be a Pareto-improvement available: an allocation  $y = (y_B^H, y_D^H, y_B^M, y_D^M)$  that strictly helps one of Homer or Marge (say Marge), while leaving the other (Homer) at least as well off as at  $x$
- since consumers optimize in equilibrium, Marge would have chosen this better allocation if she could afford it, so it must be strictly unaffordable; for Homer, his part of  $y$  must be weakly unaffordable
  - if it cost strictly less, he could have bought it and (by non-satiation) used the extra cash to get something even better

## Proof Cont'd

- let  $\omega_B^H, \omega_D^H, \omega_B^M, \omega_D^M$  be the endowment quantities (Homer-beers, Homer-donuts, Marge-beers, Marge-donuts)
- so  $p_B y_B^H + p_D y_D^H \geq p_B \omega_B^H + p_D \omega_D^H$ , and  
 $p_B y_B^M + p_D y_D^M > p_B \omega_B^M + p_D \omega_D^M$
- adding up,  
 $p_B (y_B^H + y_B^M) + p_D (y_D^H + y_D^M) > p_B (\omega_B^H + \omega_B^M) + p_D (\omega_D^H + \omega_D^M)$
- so  $p_B (y_B^H + y_B^M - \text{beer supply}) + p_D (y_D^H + y_D^M - \text{donut supply}) > 0$
- thus either total beer demand exceeds supply, or total donut demand exceeds supply
- contradicting feasibility!
- QED

# Converse?

- so equilibrium  $\Rightarrow$  Pareto-efficient
- is the converse true? Is a Pareto-efficient allocation necessarily an equilibrium?
- the answer is *partially yes*:

## Theorem

*(2nd Welfare Theorem): If preferences are convex, then any Pareto-efficient allocation can be turned into a competitive equilibrium, for a suitably chosen price vector and distribution of endowments.*

## Illustration: 2nd Welfare Theorem

- the caveat says that to turn a Pareto optimal allocation into an equilibrium (a market-clearing allocation they'd choose for themselves), you need to split the initial endowment appropriately
- reason: in our example above, one Pareto optimal allocations is to give Homer everything (16 beers and 8 donuts), and Marge nothing
- but of course, we cannot make this an equilibrium unless Homer is already endowed with everything
- for if Marge starts off with part of the endowment, it would violate the \*optimization\* part of equilibrium if she moved from this to nothing
- but it is an equilibrium if we give Homer the entire endowment and pick prices appropriately

# Summary

- so any equilibrium is Pareto-optimal (under the very weak condition of locally non-satiated preferences)
- whereas a Pareto-optimal allocation can be turned into an equilibrium for some, but not all, initial splits of the aggregate endowment
- intuitively, this says:
  - the free market will find an efficient allocation: this is NOT true with externalities! (later)
  - the only reason the government might intervene is for redistribution goals
  - any allocation they want, can be achieved via endowment taxes/subsidies